Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/08/2009
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Characteristics of Light Exposure Necessary for Development of Optimal Countermeasures to Facilitate Circadian Adaptation and Enhance Alertness and Cognitive Performance in Space		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	slockley@hms.harvard.edu	**Fax:**	FY 617 732-4015
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-4977
Organization Name:	Brigham and Women's Hospital		
PI Address 1:	Division of Sleep Medicine and Circadian Disorders		
PI Address 2:	221 Longwood Avenue, Suite 438		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	10/01/2007	**End Date:**	09/30/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aeschbach, Daniel (Brigham and Women's Hospital) Gooley, Joshua (Brigham and Women's Hospital) Ronda, Joseph (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HPF01301		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space missions often expose crew members to unusual light-dark cycles that lead to misalignment between the circadian pacemaker and the sleep-wake schedule, resulting in disturbed sleep and impaired waking function. Sleep disturbances due to other reasons (e.g., noise, temperature, microgravity) further exacerbate cognitive deficits and these factors jointly increase the risk of fatigue-related accidents and injuries. An effective countermeasure is required to facilitate more rapid adaptation of the circadian system and directly enhance alertness and performance. Light exposure induces both of these effects and is an effective, safe, non-pharmacological countermeasure for circadian- and fatigue-related deficits in cognition. Recently, we and others have shown that short-wavelength (blue) light exposure at night is the most effective wavelength for phase-shifting the circadian pacemaker, suppressing pineal melatonin, enhancing subjective alertness, improving performance and inducing EEG-derived brain activation that indicates a more alert state. These effects, however, have only been examined during the night using relatively bright light and recent evidence suggests that the spectral sensitivity of the circadian photoreception system may depend on the intensity and duration of light and that there may be changes in spectral sensitivity during the day. While blue light therapy to facilitate circadian adaptation and alertness has been proposed based largely on the effects of melatonin suppression at night, these data may not be transferable to the effects at other times of day (when melatonin is not produced) or for other effects of light. Before blue light is operationalized, three fundamental questions remain to be determined: 1) Does the spectral sensitivity depend on the intensity of light? 2) Do all non-visual effects of light have the same spectral sensitivity? 3) Is the spectral sensitivity of the circadian photoreception system different between the day and the night? Failure to address these questions may result in deployment of suboptimal or less energy-efficient lighting schedules. The proposed project will test whether the spectral sensitivity of the circadian photoreception system is intensity-dependent, response-dependent or time-of-day dependent, by comparing the effects of exposure to monochromatic light during the day and the night, at both high and low intensities, on circadian phase resetting and enhancement of alertness and performance. Experiment 1 aims to compare the melatonin suppression, phase resetting and alerting effects of exposure to dim 460 nm and 555 nm monochromatic light exposure at night, and compare these data to previously collected findings for bright light exposure at night using the same protocol. Data collection for Experiment 1 is complete and we are awaiting final melatonin assay results. We anticipate completing the analysis for Experiment 1 by the end of 2009. Experiment 2 will test the effects of exposure to bright 460 nm and 555 nm monochromatic light during the daytime and data collection will begin in October 2009 and is due for completion in May 2010. Experiment 3 will test the effects of exposure to dim 460 nm and 555 nm monochromatic light during the daytime and will begin on completion of Experiment 2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive, non-pharmacological countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments (space flight and semi-permanent bases) to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results may not apply to lower intensities of light (that are routinely experienced during space light) or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the intensity- and temporal-dependent characteristics of the circadian photoreception system. The current study will investigate these characteristics in a controlled environment with the long-term view to developing specialized lighting schedules to maximize alertness and performance of space crew in an environment where a even a small fatigue-related errors could have catastrophic consequences.
Task Progress:	We proposed to complete studies of 12 subjects per year, or 24 subjects by the end of Year 2 of the grant. To date, we have initiated studies in 23 subjects; 18 subjects completed the 9-day inpatient protocol and 5 subjects withdrew. We had some technical problems with the monochromatic light system required to generate the controlled lighting for the study which delayed recruitment temporarily. We have since taken extra steps to reduce future delays by purchasing a new monochromatic light generation device using other federal funds. With a new and more reliable lighting device, we anticipate meeting our recruitment goal of 36 subjects by the end of grant Year 3, at a rate of 1.5 completed subjects per month over the next 12 months. The data from experiment 1 are currently being analyzed and we anticipate completing these analyses on the primary outcomes (melatonin suppression, circadian phase resetting, subjective and objective correlates of alertness, auditory vigilance) by the end of 2009. Experiment 2 will test the effects of exposure to bright 460 nm and 555 nm monochromatic light during the daytime and data collection will begin in October 2009 and is due for completion in May 2010. Experiment 3 will test the effects of exposure to dim 460 nm and 555 nm monochromatic light during the daytime and will begin on completion of Experiment 2.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Anderson JL, Glod CA, Dai J, Cao Y, Lockley SW. "Lux vs. wavelength in light treatment of Seasonal Affective Disorder." Acta Psychiatrica Scandinavica, 2009 Sep;120(3):203-12. http://dx.doi.org/10.1111/j.1600-0447.2009.01345.x ; PMID: 19207131 , Sep-2009
Awards	Lockley SW. "2009 Harvard Club of Australia Foundation Harvard-Australia Fellowship, January 2009." Jan-2009
Awards	Lockley SW. "Member, Sleep Risk Mitigation Analysis Tool (RMAT) Panel, Human Research Program, NASA Johnson Space Center, July 2009." Jul-2009
Awards	Lockley SW. "Organizing Committee, American Society for Photobiology Topical Symposium, Thomas Jefferson University, Philadelphia, July 2009." Jul-2009
Books/Book Chapters	Lockley SW. "Circadian rhythms: Influence of light in humans." in "Encyclopedia of Neuroscience." Ed. L.R. Squire. Amsterdam ; Boston : Elsevier Academic Press, c2009. volume 5, p. 971-986. http://dx.doi.org/10.1016/B978-008045046-9.01619-3 , Aug-2009